

COUNTING HALLUCINATIONS IN DIFFUSION MODELS

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ABSTRACT

Diffusion probabilistic models (DPMs) have driven remarkable progress in generative tasks such as image and video synthesis. Yet they routinely produce hallucinated samples (*hallucinations*) that conflict with real-world knowledge, such as an implausible duplicate cup floating beside another cup. This problem persists precisely because the field lacks any feasible methodology for systematically quantifying such hallucinations, a gap that stalls progress and obscures the pathways toward next-generation generative models built under factual constraints. We close this gap. We isolate a concrete and tractable form of hallucination, which we term **counting hallucination**: the generation of an incorrect number of instances or structured objects, such as a hand image with six fingers, despite such patterns being entirely absent from the training data. To study it rigorously, we construct **CountHalluSet**, a dataset suite with well-defined counting criteria comprising ToyShape, SimObject, and RealHand. On top of these datasets we establish a standardized evaluation protocol for quantifying counting hallucinations, and use it to systematically characterize how the key sampling conditions in DPMs—solver type, ODE solver order, sampling steps, and initial noise—govern counting hallucination levels. Analyzing their relationship with common evaluation metrics such as the Fréchet Inception Distance (FID), we further show that widely-used perceptual metrics fail to capture counting hallucinations.

1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), commonly referred to as diffusion models, have become the dominant paradigm in generative modeling, producing high-quality samples across a wide range of tasks including text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). These advances notwithstanding, diffusion models still fail to faithfully capture factual properties of their training data, and they routinely generate outputs that violate consistency with real-world knowledge. A striking and recurring example is the generation of objects with an incorrect number of spatially intricate components, such as extra human fingers (Narasimhaswamy et al., 2024) or additional animal limbs (Borji, 2023).

A growing body of work now targets the identification and analysis of common failure modes in text-guided diffusion models. Borji (2023) catalogued several qualitative shortcomings of text-to-image models. Liu et al. (2024) introduced an adversarial search method to uncover undesirable behaviors and failure cases in text-to-image generation, and Wu et al. (2023) developed an iterative approach to discover and mitigate spurious correlations.

More recently, *hallucination* has emerged as a distinct and pressing failure mode in diffusion models (Aithal et al., 2024). It differs fundamentally from the text-prompt-based failure modes studied so far, which can be evaluated by measuring the alignment between input conditions and outputs (e.g., CLIP score (Hessel et al., 2021) and VQA score (Huang et al., 2025; Zarei et al., 2024)), and from obvious failure modes such as severe distortions, blur, and artifacts. Hallucinated samples (*hallucinations*) instead appear visually plausible and arise independently of the conditioning signal, manifesting as subtle violations of factual consistency with the training data—incorrect object counts, geometries that look plausible but are physically impossible, and other breaches of physical laws. These fine-grained distinctions between valid samples and hallucinations are precisely what current AI systems

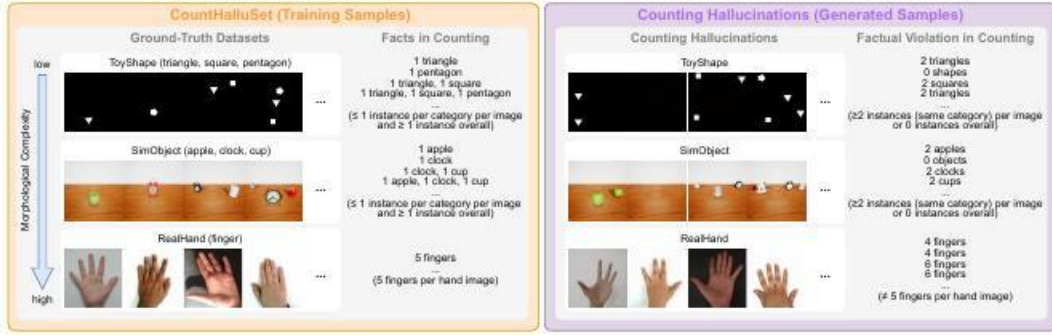


Figure 1: Comparison between the proposed **CountHalluSet** used for training (left) and the **counting hallucinations** the model generates from it (right). Each dataset defines objects under precise counting criteria. In ToyShape and SimObject, every image contains at most one instance per category and at least one instance overall; in RealHand, every image contains exactly five fingers. Counting hallucinations are generated images that break these rules—two apples in a single image, or an image with a normal background but no objects at all—patterns that never appear in the SimObject dataset.

cannot reliably detect. Even state-of-the-art large vision-language models such as GPT-4o fail at a task as basic as accurately counting the fingers in a clear hand image. A standardized, well-defined protocol for reliably quantifying hallucinated samples is therefore not merely useful but essential: it is the foundation on which next-generation generative models with strict factual consistency must be built—a capability indispensable to their role as world models (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

Among the common types of hallucinated elements—counts, colors, spatial relationships, geometry, and physical realism—counting errors are the ideal starting point for systematic study: they are both easy to identify and straightforward to quantify. We therefore define a concrete form of hallucination that we term **counting hallucination**. *Here the number of objects in the training data is explicitly predetermined, so a generated image is hallucinated whenever its object counts deviate from these predefined values.* An image depicting a hand with six fingers, for instance, is a counting hallucination, because such a pattern never occurs in the RealHand dataset (last row of Fig. 1). To enable a rigorous study of this phenomenon, we construct the CountHalluSet suite of three datasets spanning a deliberate spectrum of morphological complexity for the countable objects: ToyShape (triangle, square, pentagon), SimObject (mug, apple, clock), and RealHand (finger). Figure 1 shows representative examples from these datasets alongside their corresponding counting hallucinations.

The critical step in our quantification protocol is the accurate identification and counting of objects, which we accomplish with counting models tailored to each dataset. Equipped with these datasets and counting models, we systematically characterize how the denoising conditions of diffusion models—sampling steps, initial noise, ODE solver order, and solver type—drive counting hallucination rates. We then conduct a correlation analysis linking the counting hallucination rate to perceptual metrics, including the Fréchet Inception Distance (FID) Heusel et al. (2017) and MUSIQ Ke et al. (2021).

Our study delivers the following contributions:

- We demonstrate that commonly applied numerical strategies in diffusion models, such as increasing sampling steps from 25 to 100, fail to mitigate counting hallucinations when generating human hands, even though the same strategies are effective on simpler synthetic datasets.
- We establish that the counting hallucination rate correlates only weakly with perceptual metrics, whereas the non-counting failure rate correlates strongly with them—demonstrating that perceptual metrics cannot serve as a proxy for counting factuality.
- Building directly on these findings, we introduce joint-diffusion models, a novel training paradigm that substantially reduces both counting hallucinations and non-counting failures on the RealHand dataset.

2 RELATED WORK

2.1 HALLUCINATIONS IN DIFFUSION MODELS

Mirroring the growing attention to hallucination in large language models (LLMs) (Kadavath et al., 2022; Ji et al., 2023; Huang et al., 2023), hallucinations in unconditional diffusion models have drawn increasing interest. Aithal et al. (2024) proposed a qualitative hypothesis that hallucinated samples arise from data interpolating between distinct modes. Valuable as this insight into potential causes is, a systematic *quantitative* analysis of hallucinations in diffusion models has been entirely missing—and it is exactly this gap we address.

Our goal is to enable the accurate identification and quantification of hallucinations in diffusion-generated outputs. Such an analysis deepens our understanding of the undesirable behaviors these models exhibit and reveals how the magnitude of denoising errors—variations in sampling steps or the order of ODE solvers—controls the severity of hallucinations.

2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

The Visual Question Answering (VQA) score (Huang et al., 2025; Zarei et al., 2024) and the CLIP score (Hessel et al., 2021) are the two most widely used metrics for evaluating text-guided generated images. Yet these language-model-based evaluations are inherently ill-suited to assessing hallucinated images. There is a statistical lower bound on the propensity of pretrained language models to hallucinate certain fact types, and this bound directly caps the detection accuracy achievable by language models alone (Kalai & Vempala, 2024). The persistent performance gap between human annotators and LLMs in classifying hallucinated textual content (Lin et al., 2022; Li et al., 2023) confirms this limitation.

Image generation additionally relies on several standard quantitative metrics—FID (Heusel et al., 2017), IS (Salimans et al., 2016), Precision & Recall (Sajjadi et al., 2018), and MUSIQ (Ke et al., 2021)—all built on feature embeddings from a pretrained Inception-v3 model (Szegedy et al., 2016). Whether *any* of these metrics can capture the severity of hallucinations in generated datasets has remained an open question. We answer it directly from the perspective of counting hallucinations.

3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

3.1 FORWARD (DIFFUSION) PROCESS

Forward Process. Given the observed data $x_0 \sim q_0(x_0)$, Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) define a *forward process* (a.k.a. *diffusion process*) that gradually perturbs the data by adding Gaussian noise over T discrete timesteps. At each timestep t , the marginal distribution of the noisy sample x_t conditioned on the original data x_0 is:

$$q_{0:t}(x_t|x_0) = \mathcal{N}(x_t | \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)\mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

where $t \in \{0, 1, \dots, T\}$ and $\{\beta_i\}_{i=1}^T$ is a pre-defined, t -dependent noise schedule, typically linear (Ho et al., 2020) or cosine (Nichol & Dhariwal, 2021). The factor $\bar{\alpha}_t$ controls the signal-to-noise ratio at step t : $\bar{\alpha}_0 = 1$ (no noise) and $\bar{\alpha}_T \approx 0$ (nearly pure noise). An equivalent formulation of Eq. (1) is:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + (1 - \bar{\alpha}_t)\epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

For sufficiently large T , the marginal distribution $q_{0:T}(x_T|x_0)$ converges to a standard Gaussian $\mathcal{N}(\mathbf{0}, \mathbf{I})$. In widely-used DPMs (Ho et al., 2020; Nichol & Dhariwal, 2021), T is set to 1000.

3.2 REVERSE (DENOISING) PROCESS.

Training Objective. The *reverse process* (a.k.a. *denoising process*) of DPMs iteratively recovers the observed data from the diffused data x_T . Concretely, DPMs predict the noise ϵ from the data

$\mathbf{x}_t$ using a neural network $\epsilon_\theta(\mathbf{x}_t, t)$, the *noise prediction model*. Its parameter θ is optimized by minimizing (Ho et al., 2020; Song et al., 2022):

$$\min_{\theta} \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\pi(t) \|\epsilon_\theta(\mathbf{x}_t, t) - \epsilon\|_2^2], \quad (3)$$

where $\pi(t) > 0$ is a weighting function.

Ancestral Sampling. Given noisy data $\mathbf{x}_T$, we recover samples from the original distribution $q_0(\mathbf{x}_0)$ by reversing the diffusion process over the same number of steps, a procedure known as *ancestral sampling*:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \frac{1}{\sqrt{1-\beta_t}}(\mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)), \beta_t \mathbf{I}), \quad (4)$$

where $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ is the only unknown term, the *score function* (Song et al., 2022).

DPMs approximate the scaled score function $-\sqrt{(1-\bar{\alpha}_t)} \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ with the noise prediction model, yielding the equivalent formulation of Eq. (3):

$$\mathbf{x}_{t-1} = \mu_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon, \quad \mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1-\beta_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t). \quad (5)$$

ODE-Based Accelerated Sampling. The ancestral sampling in Eq. (4) contains a stochastic term (Kloeden et al., 1992). Its associated deterministic process is an ordinary differential equation (ODE), the *probability flow ODE* (Song et al., 2022):

$$\frac{d\mathbf{x}(\tau)}{d\tau} = f(\tau)\mathbf{x}(\tau) + h(\mathbf{x}(\tau), \tau), \quad f(\tau) = -\frac{1}{2}\beta(\tau), \quad h(\mathbf{x}(\tau), \tau) = -\frac{1}{2}\beta(\tau) \nabla_{\mathbf{x}(\tau)} \log q_\tau(\mathbf{x}(\tau)), \quad (6)$$

where $\tau \in [T, 0]$. Let $\mathbf{x}(\tau)$ denote the ideal trajectory that satisfies this ODE with the true score function $h(\mathbf{x}(\tau), \tau)$ and starts from an ideal diffused data $\mathbf{x}(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}(T))$. Set the discrete time steps as $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, where $\tau_N = 0$. Applying the variation-of-constants formula for the exact solver over one step:

$$\mathbf{x}(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h(\mathbf{x}(u), u) du \quad (7)$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear part $\frac{dz}{d\tau} = f(\tau)z$. $\Delta t > 0$ denotes the step size, which lets ODE-based solvers accelerate sampling with larger steps than ancestral sampling; the total number of sampling steps is $T/\Delta t$. Because the linear part $\mathcal{G}(t_2, t_1)$ admits a closed form, most accelerated ODE-based samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022, 2023) direct their numerical effort—higher solver orders and more sampling steps—at accurately approximating the nonlinear integral term involving h . We provide the full theoretical analysis in Appendix A.1 to A.3.

4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

This section presents our methodology and datasets for quantifying counting hallucinations. We first formalize counting criteria and the associated counting hallucinations, then introduce the datasets of the CountHalluSet suite. For each dataset we specify its counting criteria and corresponding counting hallucinations, and define the standardized protocol we use to quantify them.

4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

We give a general definition of counting criteria within training datasets and define counting hallucinations precisely as deviations from these predefined constraints.

Counting criteria in a training dataset. Let $\mathcal{D}_{\text{ref}}$ be a training dataset. For any reference sample $\mathbf{x} \in \mathcal{D}_{\text{ref}}$, the number of objects of category c in $\mathbf{x}$, $N_c(\mathbf{x})$, are given by a predefined set $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$ (e.g., $\{0, 1\}$ or $\{1, 2, 3\}$), i.e., $N_c(\mathbf{x}) \in \mathcal{S}_c$. We further require that a valid reference sample contains at least one object, i.e. $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$, where C is the set of considered categories.

Counting hallucinations as deviations from the counting criteria. Given a generated sample $\hat{\mathbf{x}}$, we say $\hat{\mathbf{x}}$ exhibits a *counting hallucination* with respect to $\mathcal{D}_{\text{ref}}$ if both of the following conditions hold:

1. **Sufficient visual quality for counting.** The image $\hat{x}$ must have sufficient visual quality that the background and any present objects are clearly discernible. This requirement cleanly separates counting hallucinations from other prominent failure modes such as severe degradations. We define a binary indicator $\mathbb{I}_{\text{CRI}}(\hat{x}) \in \{0, 1\}$, the *counting-ready indicator*, where 1 denotes image quality sufficient for counting and 0 otherwise. This indicator can be obtained via human annotation, a classifier calibrated against human annotations, or other automated criteria.
2. **Violation of counting facts.** A violation of counting facts occurs when there exists a category c with $N_c(\hat{x}) \notin \mathcal{S}_c$, or when the image contains no objects (i.e., $\sum_{c \in C} N_c(\hat{x}) = 0$), since the dataset requires at least one object per sample. We use a counting model to predict object counts in generated images as the evidence for such violations.

A counting hallucination (CH) is identified if its quality is sufficient for counting ($\mathbb{I}_{\text{CRI}}(\hat{x}) = 1$) and the predicted counts deviate from the counting criteria ($\exists c : N_c(\hat{x}) \notin \mathcal{S}_c$ or $\sum_{c \in C} N_c(\hat{x}) = 0$). Formally,

$$\mathbb{I}_{\text{CH}}(\hat{x}) = \mathbb{I}_{\text{CRI}}(\hat{x}) \wedge \left(\exists c : N_c(\hat{x}) \notin \mathcal{S}_c \vee \sum_{c \in C} N_c(\hat{x}) = 0 \right).$$

4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

The CountHalluSet suite comprises three datasets purpose-built for studying counting hallucinations: ToyShape, SimObject, and RealHand. Examples from each dataset and their corresponding counting hallucinations appear in Fig. A1 and A2, with additional statistical summaries in Appendix A.4.

4.2.1 TOYSHAPE

Dataset description. The ToyShape dataset consists of 30,000 images containing three geometric shapes: *triangle*, *square*, *pentagon*. All shapes are white, occupy an equal area of 120 pixels, and never overlap within an image. The counting criteria require each image to contain at most one instance of each shape category and at least one shape in total, as shown in the top-left of Fig. 1. Formally, we have $C = \{\text{triangle}, \text{square}, \text{pentagon}\}$. For each class $c \in C$, we set $N_c(x) \in \mathcal{S}_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(x) \geq 1$.

Counting hallucinations. Under the counting criteria above, any image generated by the ToyShape-trained model that contains two or more shapes of the same category is a counting hallucination—for example, the two-pentagon image in the top-right region of Fig. 1. Empty images (only the black background, no shapes) are likewise counting hallucinations. See Appendix A.5 for examples.

Counting model used in ToyShape. We identify counting errors with a counting model and dispense with the counting-ready indicator here, because the dataset’s simplicity makes severely degraded images rare. Specifically, we fine-tune a ResNet-50 (He et al., 2016) on over 400k ToyShape samples in which each category appears 0–3 times ($\mathcal{S}_c = \{0, 1, 2, 3\}$), deliberately spanning the full range of plausible scenarios—both counting-valid samples and counting hallucinations—to enable accurate quantification. The resulting model attains over 99.9% counting accuracy on generated data.

Quantification of counting hallucinations. Given a diffusion model trained on ToyShape, we generate a sample set matching the training-set size (30,000 images), apply the counting model to predict the number of objects in each image, and flag every image that violates the counting criteria as a counting hallucination, as illustrated in Fig. 2.

4.2.2 SIMOBJECT

Dataset description. The SimObject dataset contains 30,000 rendered images of everyday objects across three categories—*mug*, *apple*, *clock*—each with 10 distinct intra-class variations. All objects are randomly placed on a wooden tabletop under fixed lighting, with minimal mutual occlusion if any, as shown in the middle-left of Fig. 1. We synthetically generate the dataset in Unreal Engine 5 (Epic Games, 2024), using photorealistic rendering that accurately simulates object geometry, material attributes, lighting, and reflections. The counting criteria mirror those of ToyShape (see Sec. 4.2.1). Formally, we have $C = \{\text{mug}, \text{apple}, \text{clock}\}$. For each class $c \in C$, we set $N_c(x) \in \mathcal{S}_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(x) \geq 1$.

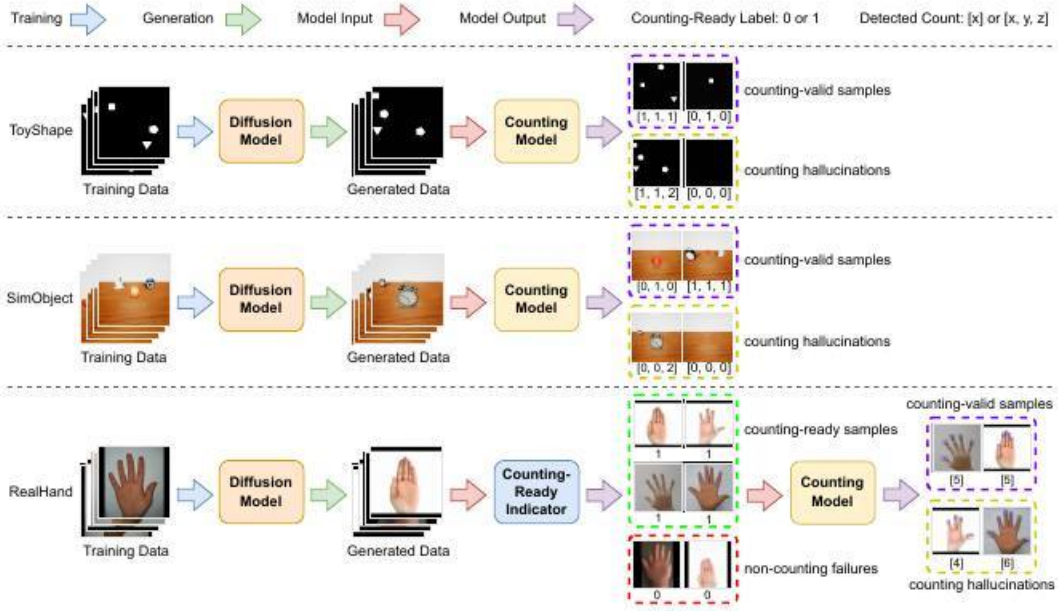


Figure 2: The evaluation procedure of the CountHalluSet suite. For ToyShape and SimObject, generated images are assessed for counting directly. For RealHand, a counting-ready indicator (a binary classifier aligned with human judgments) separates counting hallucinations from non-counting failures such as severely deformed fingers. Count labels correspond to ToyShape: *triangle*, *square*, *pentagon*; SimObject: *mug*, *apple*, *clock*; RealHand: *finger*.

Quantification of counting hallucinations. The definition of counting hallucinations, the construction of the counting model, and the quantification protocol follow exactly the ToyShape procedure, given the shared counting criteria, as shown in Fig. 2.

4.2.3 REALHAND

Dataset description. To investigate counting hallucinations in real-world scenarios, we construct the RealHand dataset of 5,050 human hand images sourced from the 11k Hands dataset (Afifi, 2019), Kaggle¹, and Roboflow². We aggregate the raw images and discard any exhibiting blur or artifacts. Each image depicts a human hand with five fingers, covering both dorsal (back) and palmar (front) views under varying lighting and backgrounds (e.g., white surfaces, blackboards, walls), as shown in the lower-left of Fig. 1. The counting criteria require each image to contain exactly five fingers, i.e., $C = \{finger\}$ with $N_c(x) \in S_c = 5$ for each $c \in C$, and $\sum_{c \in C} N_c(x) = 5$.

Counting hallucinations and non-counting failures. Generating realistic hand images inevitably produces failures beyond counting hallucinations (e.g., extra or missing fingers), including blurriness, severely deformed fingers, and visual artifacts. We isolate these cases with a counting-ready indicator (a binary classifier). Appendix A.7 provides examples of counting hallucinations and non-counting failure samples.

Counting-ready indicator and counting model used in RealHand. To separate counting hallucinations from the failure modes above, we finetune a MaxViT (Tu et al., 2022) model on 2.5k annotated generated images as a counting-ready indicator aligned with human judgments. For counting, we finetune a YOLO-12 (Tian et al., 2025) model on 2k fingertip-annotated generated images. Against human evaluation of 500 randomly sampled images by three annotators, the indicator and counting model achieve over 96% and 99% accuracy, respectively. Appendix A.6 shows example YOLO detections.

Quantification of counting hallucinations. Whereas ToyShape and SimObject allow direct evaluation of every generated image—their simplicity makes severe failures (e.g., distorted or unrecognized

¹<https://www.kaggle.com/>

²<https://universe.roboflow.com/>

able objects) rare—RealHand demands an additional filtering step. We first apply the counting-ready indicator to retain hand images of sufficient visual quality, then apply the counting model to predict the number of fingers, as shown in Figure 2.

5 EXPERIMENTS

5.1 IMPLEMENTATION DETAILS

Training of Diffusion Models. We train a DDPM (Ho et al., 2020) from scratch on ToyShape and fine-tune latent diffusion models (LDMs) (Rombach et al., 2022), pretrained on CelebA-HQ (Karras et al., 2018), on SimObject and RealHand. We set the time step T to 1,000 with a standard linear noise schedule and resize images to 128×128 for DDPMs and 256×256 for LDMs. We optimize with Adam (Kingma, 2015) at a learning rate of 0.0001 for 150k, 300k, and 80k steps on ToyShape, SimObject, and RealHand, respectively, with an effective batch size of 256.

Inference of Diffusion Models. For ancestral sampling (i.e., DDPM), we use 1,000 sampling steps. For the ODE-based solvers, we cover both first- and second-order methods: DPM-Solver-1 (equivalent to DDIM (Song et al., 2021)) as the first-order solver and DPM-Solver-2 (Lu et al., 2022) as the second-order solver. We evaluate each solver under three widely-used denoising step configurations: 25, 50, and 100 steps. On RealHand, the counting model uses a YOLO detector with a confidence threshold of 0.3 and an IoU threshold of 0.1. Unless noted otherwise, the number of generated samples matches the number of training samples in each evaluation, and every result is averaged over three random seeds.

5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

Table 1 reports the counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) under varying denoising conditions—solver types, ODE solver orders, and initial noise settings—across ToyShape, SimObject, and RealHand. The results yield five clear findings.

(1) **Increasing sampling steps reduces counting hallucinations in synthetic datasets but aggravates them in RealHand.** Under the commonly used ODE solver settings (25, 50, and 100 steps), Table 1 shows that adding sampling steps consistently lowers counting hallucination rates on ToyShape and SimObject yet raises them on RealHand (the sole exception being 100 steps with DPM-Solver-2). This contrast is telling: the simpler, more structured distributions of synthetic datasets benefit from finer solver granularity, whereas the more complex distribution of RealHand causes additional steps to overfit local inconsistencies and thereby amplify hallucinations.

(2) **Higher-order ODE solvers mitigate non-counting failures but aggravate counting hallucinations in RealHand at standard inference steps.** Comparing DPM-Solver-1 and DPM-Solver-2 on RealHand at standard sampling steps (e.g., 25 and 50 steps), Table 1 exposes a consistent trade-off: the higher-order solver substantially reduces non-counting failures while simultaneously driving up the counting hallucination rate. The same trade-off persists as we increase the number of sampling steps for both solvers. The conclusion is unambiguous: spending more computation (“better” sampling conditions) reliably improves perceptual quality (lower non-counting failure rates) but does nothing to enforce the factual correctness of finger counts.

(3) **DDPM decisively outperforms ODE solvers at mitigating both counting hallucinations and non-counting failures.** Across every comparison, ancestral sampling (DDPM) achieves the lowest CHR, NCFR, and TFR, establishing DDPM as a practical lower bound on the failure rates of generative models. Ancestral sampling is thus the most effective strategy we identify for minimizing hallucinations.

(4) **A better initial noise configuration reduces both counting hallucinations and non-counting failures.** As Table 1 shows, “diffused” noise (ground-truth initial noise) yields lower CHR, NCFR, and TFR than “normal” noise (standard Gaussian noise). The reason is that diffused noise better matches the initial noise distribution seen during training; a more informed initialization therefore mitigates not only counting hallucinations but also other non-counting failures.

Table 1: Counting hallucination rates (CHR; counting hallucinations divided by total generated samples) across different solver configurations and initial noise settings on three datasets: ToyShape, SimObject, RealHand. For RealHand we additionally report the non-counting failure rate (NCFR), the fraction of failure cases captured by the counting-ready indicator. The total failure rate (TFR) is the sum of CHR and NCFR. *Diffused* and *Normal* denote the ground-truth initial noise and standard Gaussian noise, respectively.

Solver Name	Sampling Steps	CHR (ToyShape)		CHR (SimObject)		CHR (RealHand)		NCFR (RealHand)		TFR (RealHand)	
		<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>
DPM-Solver-1 (Song et al., 2021)	25	2.43	3.47	9.27	9.95	12.71	12.95	16.18	18.06	28.90	31.01
	50	1.83	2.79	8.53	9.27	13.82	13.85	11.32	12.51	25.14	26.36
	100	1.56	2.32	8.04	8.90	14.29	14.55	9.54	10.63	23.83	25.19
DPM-Solver-2 (Lu et al., 2022)	25	2.45	3.81	8.16	8.19	14.23	14.48	8.37	9.33	22.60	23.82
	50	1.76	2.86	7.90	7.95	16.15	15.99	6.38	7.22	22.53	23.21
	100	1.41	2.24	7.89	7.89	15.06	15.43	7.82	8.94	22.88	24.38
DDPM (Ho et al., 2020)	1000	0.63	0.64	4.98	4.99	10.82	10.75	2.07	2.39	12.88	13.14

(5) Greater morphological complexity in countable objects increases counting hallucinations.

Comparing CHR across ToyShape, SimObject, and RealHand, the counting hallucination rate rises monotonically with the morphological complexity of the target objects. The fine-grained details and complex spatial relationships of morphologically complex objects make it markedly harder for diffusion models to maintain accurate object counts, producing more frequent additions and omissions.

5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

Because counting hallucination is a form of factualness hallucination, a natural question arises: *does a better perceptual metric, such as a lower FID, necessarily imply fewer counting hallucinations by reflecting a smaller distribution gap between generated and training data?* Our answer is an emphatic **no**.

To settle this question, we correlate CHR with perceptual metrics and, in parallel, NCFR with the same metrics. We use two widely used perceptual measures: a distribution-based metric, FID (Heusel et al., 2017), and an instance-based metric, MUSIQ (Ke et al., 2021). We estimate correlations with three complementary measures that together cover the full spectrum of dependence: Pearson correlation (Lee Rodgers & Nicewander, 1988) (linear), Spearman rank correlation (Spearman, 1904) (monotonic), and Distance Correlation (Székely et al., 2007) (general nonlinear). The evidence below establishes our conclusion.

👉 **The correlation between CHR and perceptual metrics is dataset- and solver-dependent, not intrinsic.** The first and third rows of Table 2 show that the Pearson correlation between CHR and FID swings across datasets—positive for SimObject, negative for RealHand, positive again for PTI-Hand—while the corresponding Spearman correlations move from positive to negative to negative. The distance correlations fall from 0.82 (SimObject) to 0.74 (RealHand) to 0.47 (PTI-Hand), demonstrating that more complex generative content progressively erodes any association between CHR and FID. Integrating DDPM results with those from ODE solvers attenuates these correlations still further.

👉 **The correlation between NCFR and perceptual metrics is uniformly strong and positive.** In sharp contrast to the erratic CHR correlations, the correlation between NCFR and perceptual metrics is strong and robust in every setting. For NCFR vs. FID, all coefficients exceed 0.88 with very small p-values, confirming high statistical significance.

5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

We now show that our findings extend beyond controlled settings to a large-scale pretrained text-to-image model. We evaluate Stable-Diffusion-3.5-Medium (Esser et al., 2024) with the counting-ready indicator and counting model of Sec. 4.2.3; further details appear in Appendix A.8. As Figure 3 makes clear, the way solver type and sampling steps affect counting hallucinations in this large pretrained model closely matches the trends from our controlled RealHand experiments. This agreement directly

Table 2: Correlation of the counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) with perceptual metrics FID and MUSIQ. PTI-Hand results are obtained by prompting a pretrained Stable-Diffusion-3.5-Medium model with ten hand-generation prompts to produce 5050 samples, matching the size of the RealHand dataset. FID for PTI-Hand is computed between RealHand images and the model-generated samples. Here p is the p-value assessing the statistical significance of the correlation r or ρ . † marks DDPM results integrated with those from various ODE solvers across sampling conditions.

Metric	CHR vs. FID						NCFR vs. FID						CHR vs. MUSIQ						NCFR vs. MUSIQ					
	SimObject	RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	
Pearson (r)	0.87	0.01	-0.97	<0.01	0.20	0.69	0.99	<0.01	0.88	<0.01	0.93	<0.01	-0.02	0.95	-0.97	<0.01	-0.99	<0.01						
Pearson† (r)	0.49	0.14	-0.53	0.13	—	—	0.96	<0.01	—	—	0.46	0.20	—	—	-0.98	<0.01	—	—						
Spearman (ρ)	0.84	<0.01	-0.97	<0.01	-0.14	0.78	0.99	<0.01	0.92	<0.01	0.80	0.01	0.42	0.33	-0.88	<0.01	-0.46	0.29						
Spearman† (ρ)	0.75	0.01	-0.51	0.15	—	—	0.95	<0.01	—	—	0.40	0.28	—	—	-0.91	<0.01	—	—						
Distance Correlation (r_d)	0.89	—	0.97	—	0.47	—	0.99	—	0.91	—	0.93	—	0.42	—	0.97	—	0.99	—						
Distance Correlation† (r_d)	0.82	—	0.74	—	—	—	0.95	—	—	—	0.71	—	—	—	0.97	—	—	—						

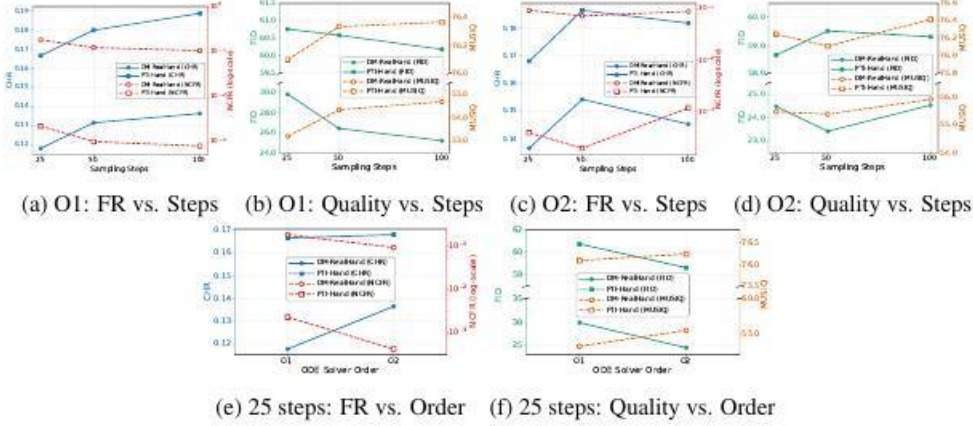


Figure 3: Trend consistency between DM-RealHand (ours) and PTI-Hand (pretrained Stable-Diffusion-3.5-Medium). O1 and O2 denote first- and second-order ODE solvers (DPM-solver-1 / DPM-solver-2 for DM-RealHand; Euler / Heun for PTI-Hand). FR metrics are CHR and NCFR; quality metrics are FID and MUSIQ. Across variations in sampling steps and solver order, the two models track each other closely, confirming that our findings transfer to large pretrained models.

validates the findings of Sec. 5.2 and confirms the broad applicability of our evaluation protocol. The correlation results in Table 2 reinforce the same conclusion: we again observe weak correlations between CHR and perceptual metrics and strong correlations between NCFR and perceptual metrics. Together, these results establish the robustness and generality of our findings and demonstrate that the proposed evaluation protocol transfers reliably to pretrained text-to-image models.

5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

The lower counting hallucination rates on simpler countable objects (Observation 5 in Sec. 5.2) point to a clear principle: diffusion models capture counting-based factuality far more reliably when the underlying visual structure has low morphological complexity (e.g., masked objects). This principle motivates a concrete question: if a diffusion model denoises general visual representations *together* with explicit, simple structural constraints that are easier to model—object masks encoding basic topology and counts—within a shared latent space, will such a joint learning paradigm suppress counting hallucinations? We answer this affirmatively and introduce **joint-diffusion models** (JDM), models that jointly learn general visual representations and structural constraints during denoising.

Concretely, we concatenate the latent representation of the hand mask channel-wise with that of the corresponding RGB image, as shown in Fig. 4. We then finetune the UNet pretrained on CelebA-HQ (Karras et al., 2018) (following the RealHand configuration of Sec. 5.1) on the RealHand images and their masks, extracted with SAM-2 (Ravi et al., 2024). To accommodate both modalities, we expand

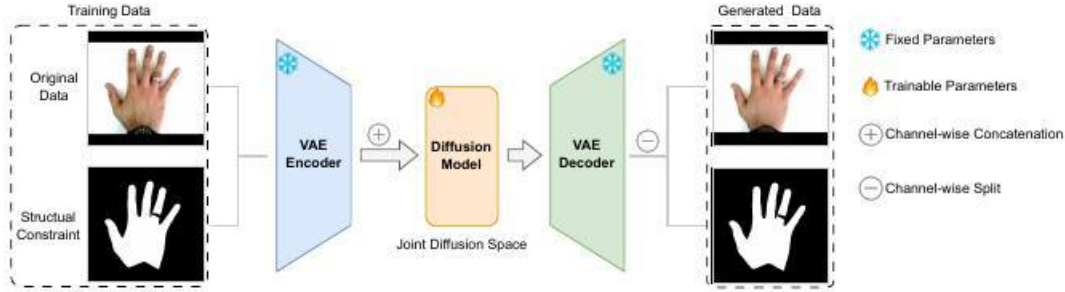


Figure 4: The proposed joint-diffusion model (JDM). JDM concatenates the latent representations of hand RGB images and their segmentation masks channel-wise. This joint learning paradigm enforces anatomically plausible hand poses and correct finger topology during generation, yielding lower CHR and NCFR.

Table 3: Failure rates of LDM versus our proposed JDM on the RealHand dataset across different ODE solvers and sampling steps. Lower is better. Although JDM jointly generates hand RGB images and segmentation masks, we compute FID solely on the RGB outputs to keep the comparison with baselines fair.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
LDM	12.95	13.85	14.55	14.48	15.99	15.43	18.06	12.51	10.63	9.33	7.22	8.94	29.86	26.43	25.21	24.48	23.42	24.54
JDM (ours)	11.41	10.67	10.51	10.12	9.66	10.07	2.94	2.42	2.29	3.51	4.63	2.26	37.69	35.51	34.68	34.63	32.07	33.89

the input and output channels of the denoising UNet to match the combined latent dimensionality of the image and mask latents (3+3=6 channels).

Table 3 confirms the paradigm’s effectiveness: across diverse sampling conditions, JDM consistently drives both CHR and NCFR below the vanilla LDM, demonstrating that it generates structurally intricate objects prone to counting ambiguities far more accurately. The one cost is a higher FID, a direct and expected consequence of jointly modeling two modalities under a small-scale training regime, which trades some perceptual quality for markedly improved structural realism. Crucially, we also observe promising reductions in counting hallucinations when applying JDM to large-scale pretrained diffusion models under carefully controlled finetuning settings. Appendix A.9 provides further details, results, and discussion.

6 CONCLUSION

This work delivers the first systematic framework for quantifying hallucinations in diffusion models and, with it, new insights into improving factual generation. Focusing on counting hallucinations, we introduce **CountHalluSet** together with a practical, reproducible methodology for measuring them. Through comprehensive and controlled experiments, we show that commonly adopted numerical strategies fail to mitigate counting hallucinations when generating human hands. Our correlation analyses further establish that widely used perceptual metrics (e.g., FID and MUSIQ) cannot consistently reflect the severity of counting hallucinations. Experiments on large-scale pretrained text-to-image models confirm both the generality and the validity of these findings, and our joint-diffusion paradigm charts a concrete path toward factually consistent generation.

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A APPENDIX

A.1 POOF OF THE BOUND OF TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

For the joint distributions of the ideal reverse process $q(\mathbf{x}_{T:0})$ and the true reverse process $p_\theta(\mathbf{x}_{T:0})$, their KL divergence is defined as:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

For both reverse processes $q(\mathbf{x}_{T:0})$ and $p_\theta(\mathbf{x}_{T:0})$, we can decompose them as follows, leveraging the Markov property:

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

By applying the chain rule to decompose the KL divergence, we have:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[\mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[\log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

Then, according to the Data Processing Inequality (Cover & Thomas, 2012), marginalizing (or discarding some random variables) will not increase the KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0}) \| p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

The bound of total accumulated distributional error can be obtained by combining Eq. (A.4) and Eq. (A.5):

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

where equality ($=$) holds strictly only in the ideal case $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$ and $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ for all t and $\mathbf{x}_t$, i.e., when the diffusion model exactly recovers the underlying data distribution $q(\mathbf{x}_0)$. This upper bound on the overall KL divergence $\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0) \| p_\theta(\mathbf{x}_0))$ between true and model-generated data is identical to the variational bound derived in DDPM (Ho et al., 2020) once the data entropy $H(\mathbf{x}_0)$ is excluded. Optimizing DDPM’s variational loss therefore directly tightens this explicit upper bound on the discrepancy between true and model-generated data.

Diffused-prior gap. The first source of error is the mismatch between the distribution of the ground-truth diffused data $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T | \mathbf{x}_0)$ and the prior distribution used for sampling $p_\theta(\mathbf{x}_T)$. This mismatch is a direct consequence of the number of diffusion steps T being finite in practice. We quantify the discrepancy with a general distributional metric:

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T) \| p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

where $\mathcal{D}_{\text{KL}}$ denotes the Kullback–Leibler (KL) divergence. Other appropriate metrics, such as FID or the Wasserstein distance, evaluate this distributional discrepancy equally well. Present from the

outset, the diffused-prior gap induces a deviation that persists throughout the denoising process and ultimately yields a sustained output error.

Noise model prediction error. The second source of error is the noise prediction model itself. Its estimate $\epsilon_\theta(\mathbf{x}_t, t)$ only approximates the true noise ϵ —the *Universal Approximation Theorem* (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022) guarantees this approximation in principle but not exactness. Let $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$ denote this prediction error. By Eq. (5), any discrepancy in ϵ_θ induces a corresponding deviation in the predicted mean at step t :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Since each reverse step relies on the mean from the previous iteration, this bias accumulates along the denoising trajectory and thereby contributes to the overall prediction error.

A.2 POOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

Building on Eq. (7), we take a one-step numerical solver as our example. Let $\mathbf{x}^*(\tau)$ denote the ideal trajectory that satisfies this ODE with the true score function $h^*(\mathbf{x}(\tau), \tau)$ and starts from a ground-truth initial condition $\mathbf{x}^*(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}_T)$. Let $\hat{\mathbf{x}}_k$ denote the trajectory computed by a numerical ODE solver at discrete time steps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, and let $\hat{h}$ be the estimated model function. We now compute the total accumulated trajectory error of a specific numerical trajectory from $\hat{\mathbf{x}}_0$ to $\hat{\mathbf{x}}_N$. Let $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$ be the trajectory error at step k , and consider the transition from τ_k to $\tau_{k+1} = \tau_k - \Delta t$.

Ideal path evolution. Using the variation-of-constants formula for the exact solution over one step (from τ_k down to τ_{k+1}):

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear part $\frac{dz}{d\tau} = f(\tau)z$. Note that the integration limits are backward in time.

Numerical path evolution. A one-step numerical solver approximates the solution, and this numerical step deviates from the exact evolution governed by $\hat{h}$ starting from $\hat{\mathbf{x}}_k$. Let $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h})$ be the exact solution at τ_{k+1} of the ODE using $\hat{h}$, starting from $\hat{\mathbf{x}}_k$ at τ_k :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The numerical method introduces a local truncation error δ_k :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) + \delta_k. \quad (\text{A.11})$$

So, we have:

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

Error recurrence. Subtract the ideal evolution Eq. (A.9) from the numerical evolution Eq. (A.12):

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

Accumulated error formulation. We can unroll this recurrence relation from $k = 0$ to $N - 1$. Let $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}) \dots \mathcal{G}(\tau_{k+1}, \tau_k)$ for $j > k$, and $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$. Applying the recurrence repeatedly:

$$\begin{aligned} e_N &= \mathcal{G}(\tau_N, \tau_0) e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \\ &\quad \cdot \left(\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \end{aligned} \quad (\text{A.14})$$

Substituting $\tau_N = 0$, $\tau_0 = T$, $e_N = \hat{\mathbf{x}}_N - \mathbf{x}^*(0)$, and $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$:

$$\begin{aligned} \hat{\mathbf{x}}_N - \mathbf{x}^*(0) = & \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \\ & \cdot \left(\underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \end{aligned} \quad (\text{A.15})$$

where the propagated initial error term captures the deterministic impact on the final path error of a specific initial error instance, whose statistical origin is the distributional diffused-prior gap.

A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

As shown in Eq. (2), the forward process in DDPM (Ho et al., 2020) can be formulated as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

And the reverse process is defined as follows:

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \frac{1 - \bar{\alpha}_t}{\sqrt{\bar{\alpha}_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon. \quad (\text{A.17})$$

For DDIM (Song et al., 2021), the forward process is identical to that in Eq. (A.16), while its deterministic reverse process (with $\eta = 0$) is defined as follows:

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_\theta(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \left(\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \epsilon_\theta(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Although DDPM and DDIM share the same forward (diffusion) process, DDIM does *not* merely strip the stochastic term from DDPM's reverse (denoising) process: the two methods employ genuinely different coefficient formulations for the predicted noise term.

A.4 DATASET EXAMPLES AND DETAILS

Figure A1 presents representative examples from the three datasets used in this study. Table A1 shows that each dataset is balanced across its categories, and Table A2 shows that the distribution of images by object count is comparable—together confirming the absence of object-count bias across datasets. Both tables list the datasets in the same order, and the category order in Table A1 is:

- **ToyShape**: triangle, square, pentagon;
- **SimObject**: mug, apple, clock;
- **RealHand**: finger.

Table A1: Number of objects per category for each dataset.

	ToyShape	SimObject	RealHand
Category 1	20,027	19,916	25,250
Category 2	19,902	20,101	-
Category 3	19,978	19,974	-

Table A2: Number of images by object count(s) per dataset.

	ToyShape	SimObject	RealHand
1 object	10,003	10,022	-
2 objects	10,087	9,965	-
3 objects	9,910	10,013	-
5 objects	-	-	25,250
Total	30,000	30,000	25,250

A.5 EXAMPLES OF COUNTING HALLUCINATIONS

Fig. A2 shows representative counting hallucinations for the three datasets. For ToyShape and SimObject, the counting hallucinations contain more than two instances of at least one category, or no instances at all against a normal background, as shown in Fig. A2(a) and Fig. A2(b). For RealHand, counting-hallucinated samples are marked by incorrect finger counts (e.g., 4 or 6 fingers), as shown in Fig. A2(c). Note that the last two images of the first row in Fig. A2(c) appear to show five fingers, but one finger is partially incomplete and visibly artifacted, leaving only four valid fingers; these samples are therefore correctly classified as counting hallucinations.

A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

Figure A3 presents representative YOLO predictions on the RealHand dataset. The YOLO detector accurately identifies fingertips across diverse hand poses and orientations, spanning both dorsal and palmar views. It does, however, miss anatomically implausible configurations such as hands with duplicated thumbs (e.g., the penultimate sub-figure of the first row and the last sub-figure of the last row in Fig. A3(a)) or hands with five middle fingers and no thumb (e.g., the last sub-figure of the first row and the penultimate sub-figure of the last row in Fig. A3(a)). Such counting-valid samples are still detected as five-finger hands and remain indistinguishable from anatomically valid ones by finger count.

Figure A3(b) shows counting-hallucinated samples where YOLO predicts incorrect fingertip counts (e.g., 3, 4, 6, or 7 fingertips). Figure A3(c) illustrates the rare edge cases (<1%) in which the detector fails on fingertips with unclear boundaries, cracked fingers, or floating artifacts. Tuning YOLO’s inference hyperparameters (e.g., IoU and confidence thresholds) partly mitigates these cases, but at the cost of detection performance on the majority of counting-valid samples. Because we apply a single consistent detection setup across all RealHand evaluations, these rare cases leave our overall conclusions intact, and addressing them further would be a challenging, low-reward effort.

A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

Figure A4 shows representative examples of the three classes of generated hand images: counting-valid, counting-hallucinated, and non-counting failure samples. As illustrated in Fig. 2, the RealHand evaluation procedure first applies the counting-ready indicator to separate counting-ready samples (counting-valid and counting-hallucinated) from images unsuitable for counting, which typically exhibit severe visual artifacts such as distorted, cracked, partially visible, or floating fingers. It then classifies each counting-ready sample as counting-valid or counting-hallucinated according to whether the detected number of valid fingertips equals five.

A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

In the PTI-Hand experiments, we prompt Stable-Diffusion-3.5-Medium with 10 distinct hand-generation prompts, listed below.

1. A single human hand, front view, all fingers in frame and visible.
2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.

4. Overhead shot of a person’s hand, entire hand in frame, all fingers visible, plain background.
5. One hand, top view, five fingers visible, simple background.
6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

Our definition of hallucination assumes the model generates patterns never observed during training. For large-scale pretrained diffusion models this assumption is hard to verify directly, since the full training distribution is inaccessible. To keep the quantification of counting hallucinations feasible in this case, we adopt a deliberately conservative criterion: we treat any generated hand image with six or more fingers as a counting-hallucinated sample. This criterion is well justified. Real images may show fewer than five visible fingers due to occlusion, articulation, or viewpoint, but real data virtually never contain hands with six or more fingers. Finger counts ≥ 6 therefore provide a reliable, empirically verifiable signal of counting hallucination. See Fig. A5 for generated examples.

A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

In the JDM experiments of Table 3, we fine-tune the denoising UNet for 120k steps, versus 80k steps for the vanilla LDM. This larger training budget is necessary because extending the latent space from three to six channels introduces a substantial mismatch with the pretrained latent distribution of LDM. The additional optimization closes this distributional gap and lets JDM’s modeling capacity fully materialize.

A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

Table A3: Failure rates and image quality for finetuning SD-1.5 versus finetuning SD-1.5 with the joint-diffusion training paradigm on RealHand. The first row (baseline) reports finetuning Stable-Diffusion-1.5 on RealHand data. The second row freezes all intermediate layers of the baseline model and finetunes only the input and output layers with extended channel dimensions. Lower is better.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fr�chet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
SD-1.5	0.3362	0.3404	0.3630	0.3396	0.3376	0.3362	0.0059	0.0032	0.0020	0.0050	0.0026	0.0018	61.51	57.71	57.70	58.53	57.57	56.69
SD-1.5 with joint-diffusion finetuning	0.2838	0.3222	0.4145	0.3053	0.3277	0.3558	0.0947	0.0511	0.0487	0.0549	0.0362	0.0327	70.65	79.22	102.39	70.80	72.84	83.07

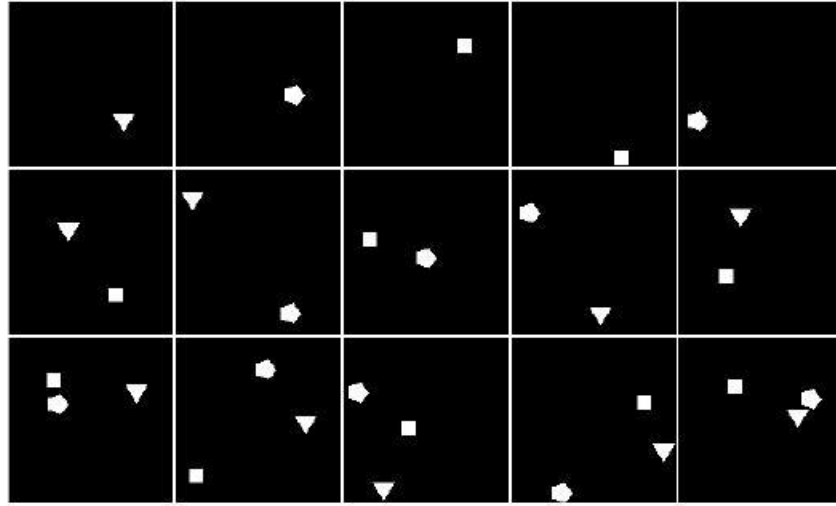
We emphasize that JDM is a **novel training paradigm** for improving factual generation in diffusion models, not a superior inference strategy. First, JDM extends only the input and output channels of the UNet and leaves every intermediate layer size unchanged, so it **introduces negligible inference-time overhead compared to vanilla LDM**. Second, extending a pretrained text-to-image model with additional mask channels **disrupts the manifold geometry of the pretrained latent space and undermines the original denoising distribution**, which makes finetuning such a model directly nontrivial.

This challenge notwithstanding, we still obtain promising reductions in counting hallucinations when applying JDM to large-scale pretrained diffusion models under carefully controlled finetuning conditions. Aligning JDM’s extended latent space with the highly rigid latent geometry of large pretrained models is difficult at the limited scale of RealHand, so we adopt a two-stage finetuning strategy: we first finetune Stable-Diffusion-1.5 on RealHand, then freeze all intermediate UNet layers and finetune only the input and output layers with extended channels. As Table A3 shows, even

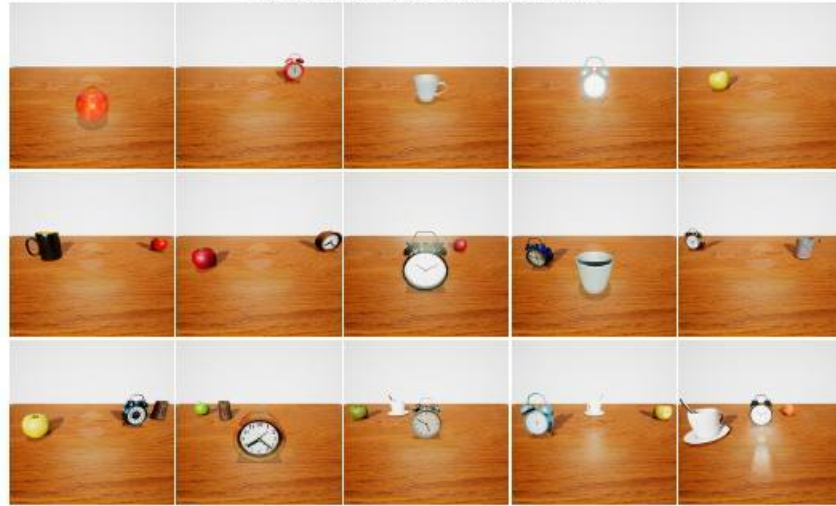
972 under this deliberately constrained setup the joint-diffusion paradigm consistently lowers counting
973 hallucination rates at 25 and 50 sampling steps across both solvers.

974
975 Taken together, this evidence establishes JDM as a promising direction for enforcing factual con-
976 sistency in image generation. **Studying its behavior and performance under true large-scale**
977 **training is an important and compelling avenue for future work.**

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(a) Examples in the ToyShape dataset

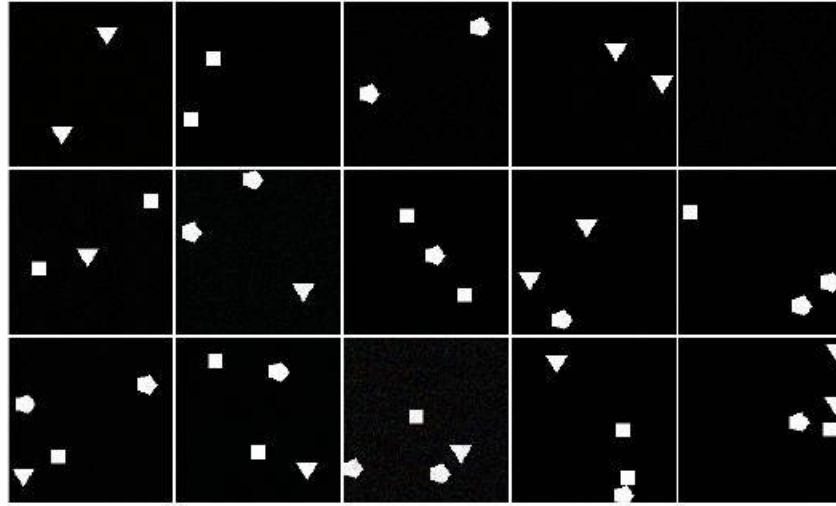


(b) Examples in the SimObject dataset



(c) Examples in the RealHand dataset

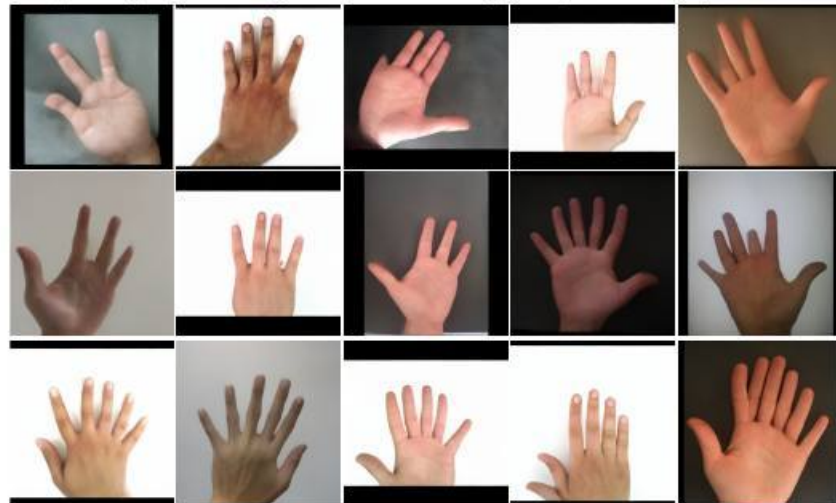
Figure A1: Representative examples from the datasets in the CountHalluSet suite: ToyShape, SimObject, and RealHand.



(a) Examples of counting hallucinations corresponding to the ToyShape dataset

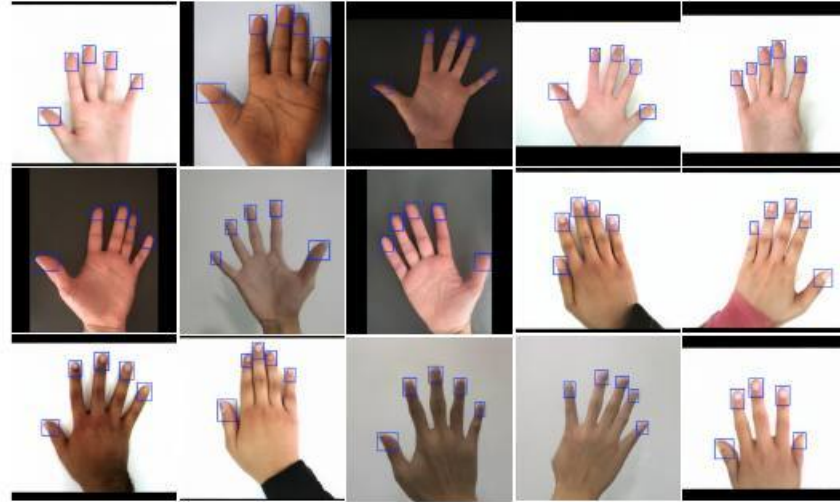


(b) Examples of counting hallucinations corresponding to the SimObject dataset

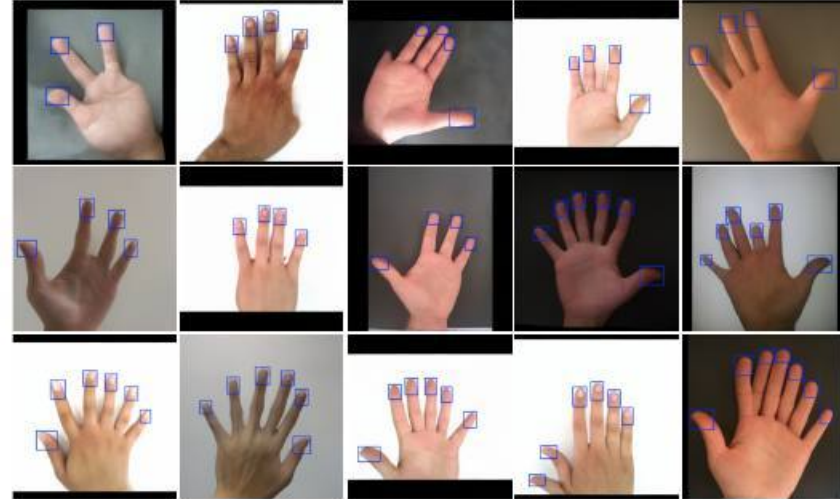


(c) Examples of counting hallucinations corresponding to the RealHand dataset

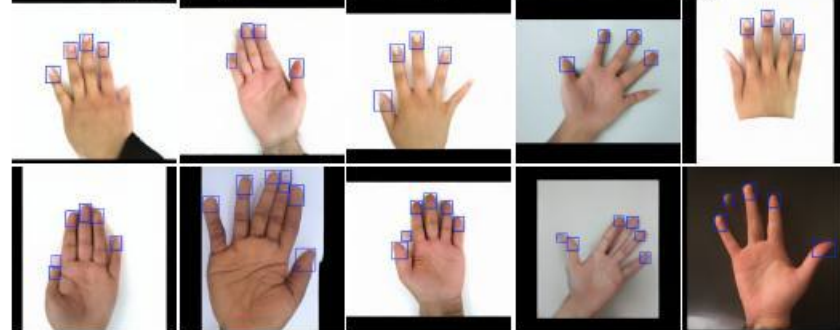
Figure A2: Examples of counting hallucinations corresponding to the ToyShape, SimObject, and RealHand datasets.



(a) Counting-valid YOLO predictions with five fingers

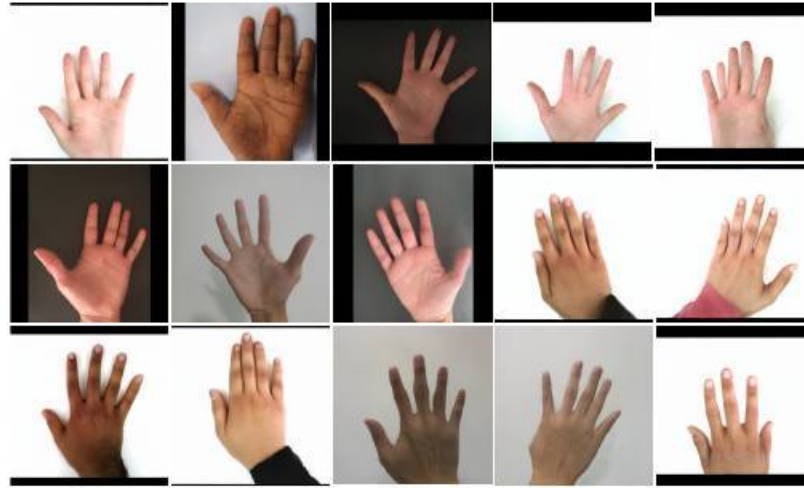


(b) Counting-hallucinated YOLO predictions with not five fingers

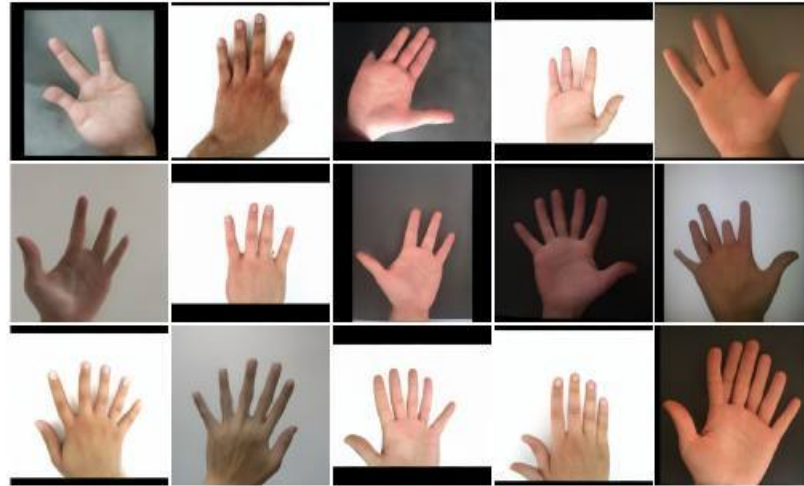


(c) Rare edge cases where YOLO predictions fail

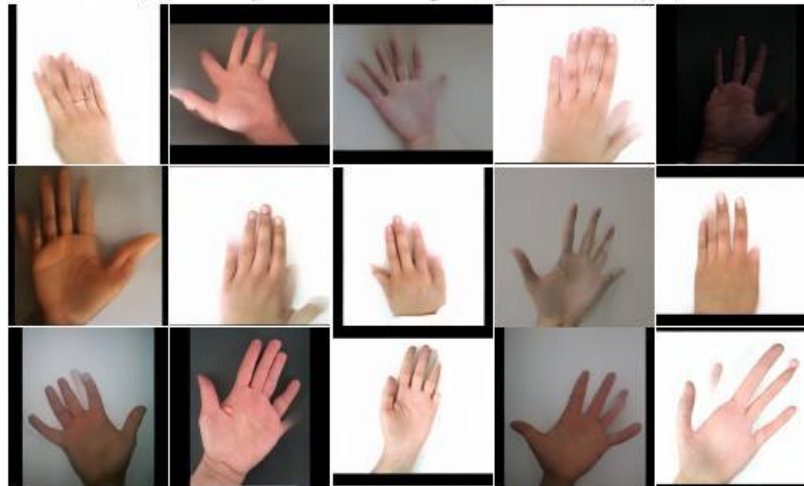
Figure A3: Examples of YOLO predictions on generated RealHand images. (a) Counting-valid samples with five fingertip bounding boxes correctly predicted by YOLO (including cases with two thumbs). (b) Counting-hallucinated samples with incorrect fingertip count predictions (e.g., 3, 4, 6, or 7 detected fingertips). (c) Several rare edge cases where the counting-ready indicator fails to filter out and YOLO struggles to handle, such as blurred, incomplete, or anatomically disconnected fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (not five valid fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A4: Examples of three classes of generated hand images. (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with an incorrect number of valid fingers (e.g., 4 or 6). (c) Non-counting failures: images unsuitable for counting due to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.

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(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (more than 5 fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A5: Examples of three classes of generated hand images using pretrained Stable-Diffusion-3.5-Medium (PTI-Hand). (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with more than five valid fingers (e.g., 6 or 7). (c) Non-counting failures: images unsuitable for counting due to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.